BRADEN NICHOLSON

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Education

University of Cape Town, Cape Town, South Africa — Jan 2022 - Oct 2026

- MScEng in Mechanical Engineering (in progress, expected submission January 2027)
  - Dissertation-based degree developing a novel extreme-temperature fracture toughness testing module
- BScEng in Mechanical Engineering (completed December 2024)
  - GPA: 78.54/100 (first class distinction, equivalent to an A)
  - SAIMechE award for best final year thesis project (grade of 95)

Northern Illinois University, DeKalb, IL, USA — Aug 2019 - Dec 2021

- BSc Mechanical Engineering (partially completed)
  - GPA: 3.701/4.0
  - Participant in Honors program throughout studies

Professional experience

Laboratory Engineering, Centre for Materials Engineering, Cape Town, South Africa — Sep 2022 - Oct 2026

- Part-time role in conjunction with studies
- Support materials research through engineering work, designing/modifying equipment to perform research
- Guide postgrad students in testing activities
- Supervise undergraduate interns, designing and overseeing their projects
- Assist with general laboratory upkeep

Lecturing, Centre for Materials Engineering, Cape Town, South Africa — Feb 2025 - Mar 2026

- Have taught a part of the Honors class "Microstructure & phase transformations in engineering metals" each year

Teaching assistant position, Department of Mechanical Engineering, University of Cape Town, Cape Town, South Africa — Jan 2025 - Dec 2025

- Assisted the mechanical engineering department's senior design class throughout 2025
- Included design guidance/teaching good practice and running workshop assembly sessions

Mechanical Engineering Internship, GasCon, Cape Town, South Africa — Jun 2024 - Jul 2024

- Interned at a large-scale pressure vessel manufacturing company
- Gained experience in engineering design of pressure vessels, in the mechanical laboratory (materials testing and validation), and the machine shop
- Streamlined the software processes in the laboratory's filing system by introducing automation of material test data creation and storage

Job Shadow Experience, Smalley Springs and Rings, Lake Zurich, IL, USA — Jan 2020

- Shadowed design engineers in Smalley, a company that designs and manufactures springs/retaining rings for a wide variety of clients
- Gained exposure to design, testing, and manufacturing, all of which was done in-house

Skills

Engineering software: SolidWorks, ANSYS Mechanical, ANSYS Electronics, Python, MATLAB, NumPy, KiCad, LTSpice, LaTeX, PlatformIO, Git
Knowledge areas: Mechanical design, mechanical and electromagnetic simulation, materials characterization, mechatronics (basic), data analysis